A Two-Phase Deep Learning Framework for Detecting Deepfake and AI-Generated Videos

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Synopsis

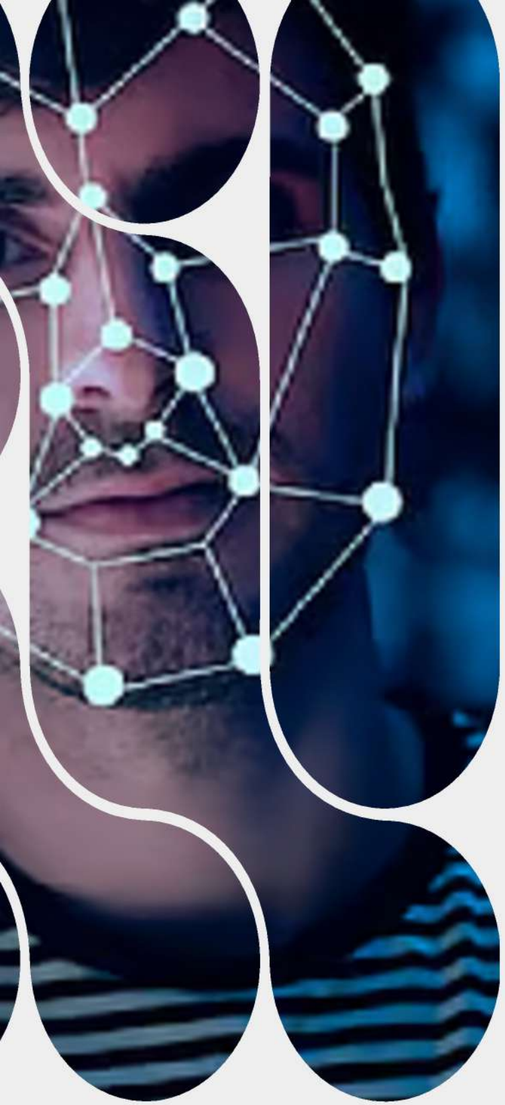
Rapid rise of AI-generated and deepfake videos threatens digital trust (e.g., misinformation, fraud).

TWO PHASE SOLUTION

- **Phase 1 detects camera-captured vs. AI-generated;**
- **Phase 2 identifies real vs. deepfake in camera videos.**

- Uses CNN ensembles (XceptionNet + EfficientNet-B2), residuals, FFT features .
- Achieves ~96-99% accuracy; robust cross-dataset generalization via merged training for deepfake manipulated videos
- Achives 99.5 % accuracy for AI Video detection

Impact: Unified forensic tool for media verification.
This thesis introduces a hierarchical detection framework to handle evolving fake media threats, outperforming single-task models



Introduction

Digital media authenticity is a major concern today
Deepfakes manipulate real videos using AI techniques
AI-generated videos are created without real-world input
Traditional systems:

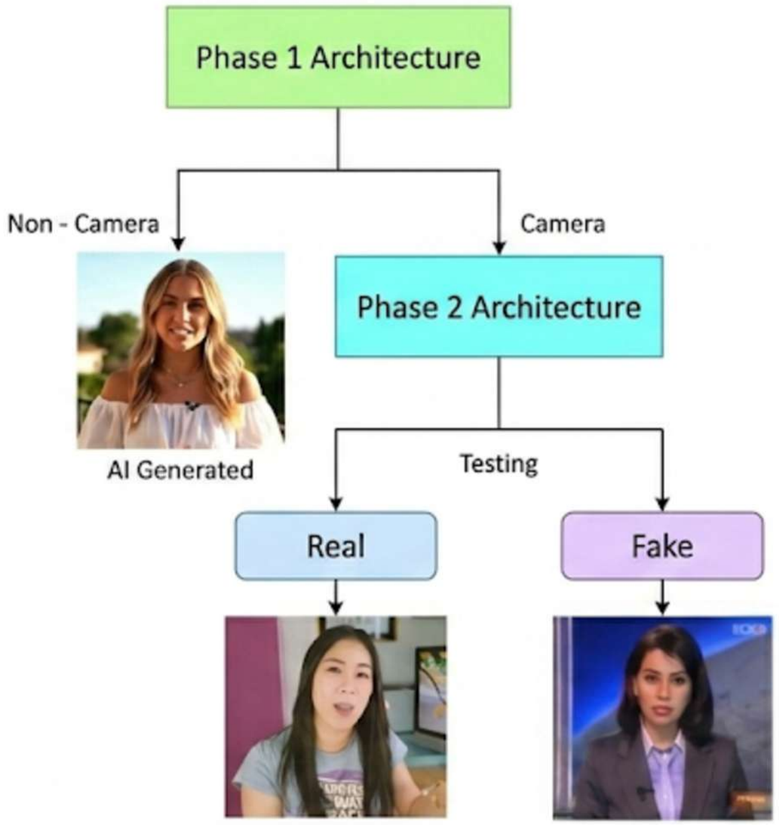
- Focus only on deepfake detection
- Ignore AI-generated content

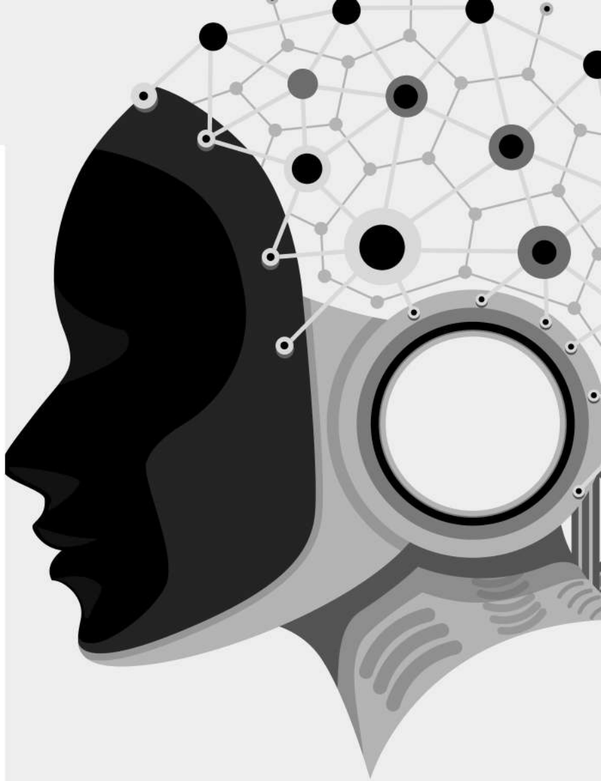
Key challenges:

- High visual similarity to real videos
- Lack of generalized models

This project proposes a unified detection framework

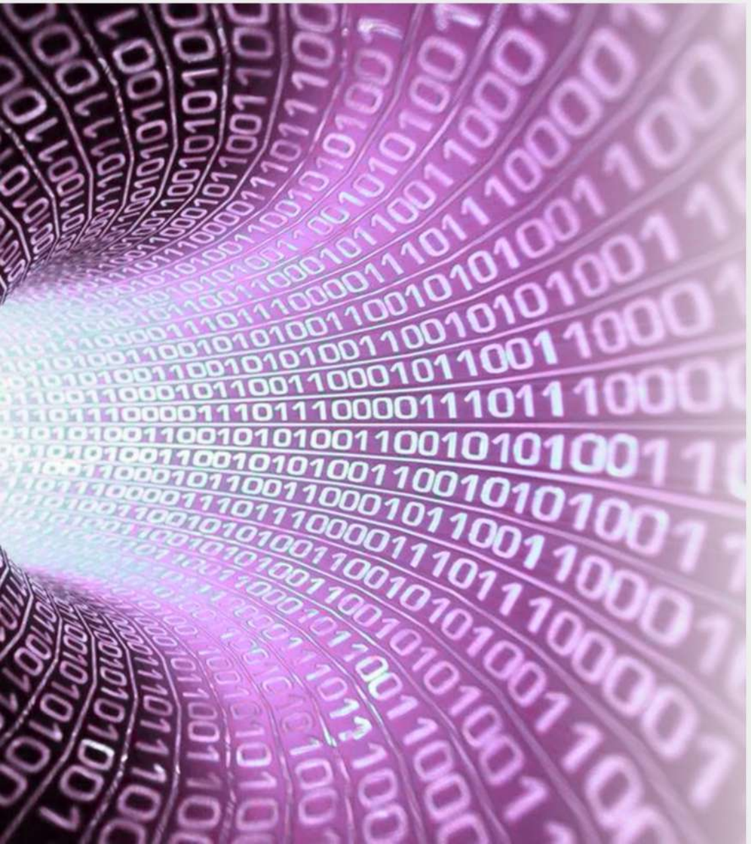
Overall Explanation





The workflow follows a two-phase pipeline for detecting forgery and identifying the source of a video. First, an input video is provided, which may be real, AI-generated, or manipulated. In Phase 1, the system performs source attribution by distinguishing whether the video originates from a camera (real-world capture) or is non-camera content such as AI-generated media. If the video is identified as non-camera, it is directly treated as synthetic. If it is classified as camera-based content, it proceeds to Phase 2, where a deeper analysis is performed to determine authenticity. In this phase, the system classifies the video as either real (genuine) or fake (manipulated or deepfake). The final output indicates both the origin and whether the video is authentic or forged.

Dataset Description



Phase 1: Dataset Preparation

- Real videos sourced from DFDC and MAVO dataset
- Synthetic videos generated using Google AI Labs and Google Cloud Vertex AI
- Balanced mix ensures diversity between real and AI-generated clips

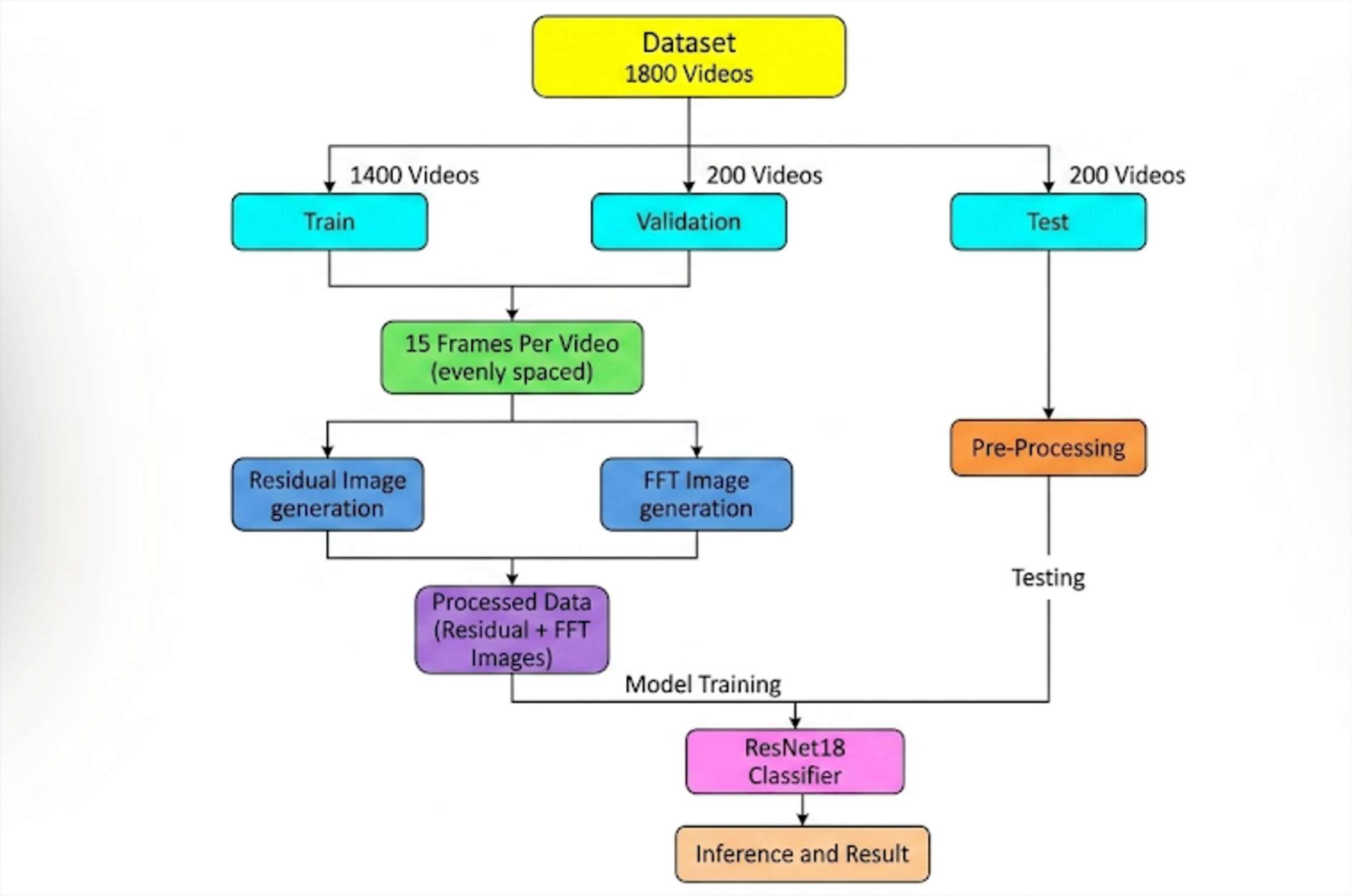
Phase 2: Multi-Dataset Learning

- Combined datasets: CelebDF, DFDC, FaceForensics++
- Includes face swaps, reenactments, and DeepFakes
- Improves cross-dataset generalization

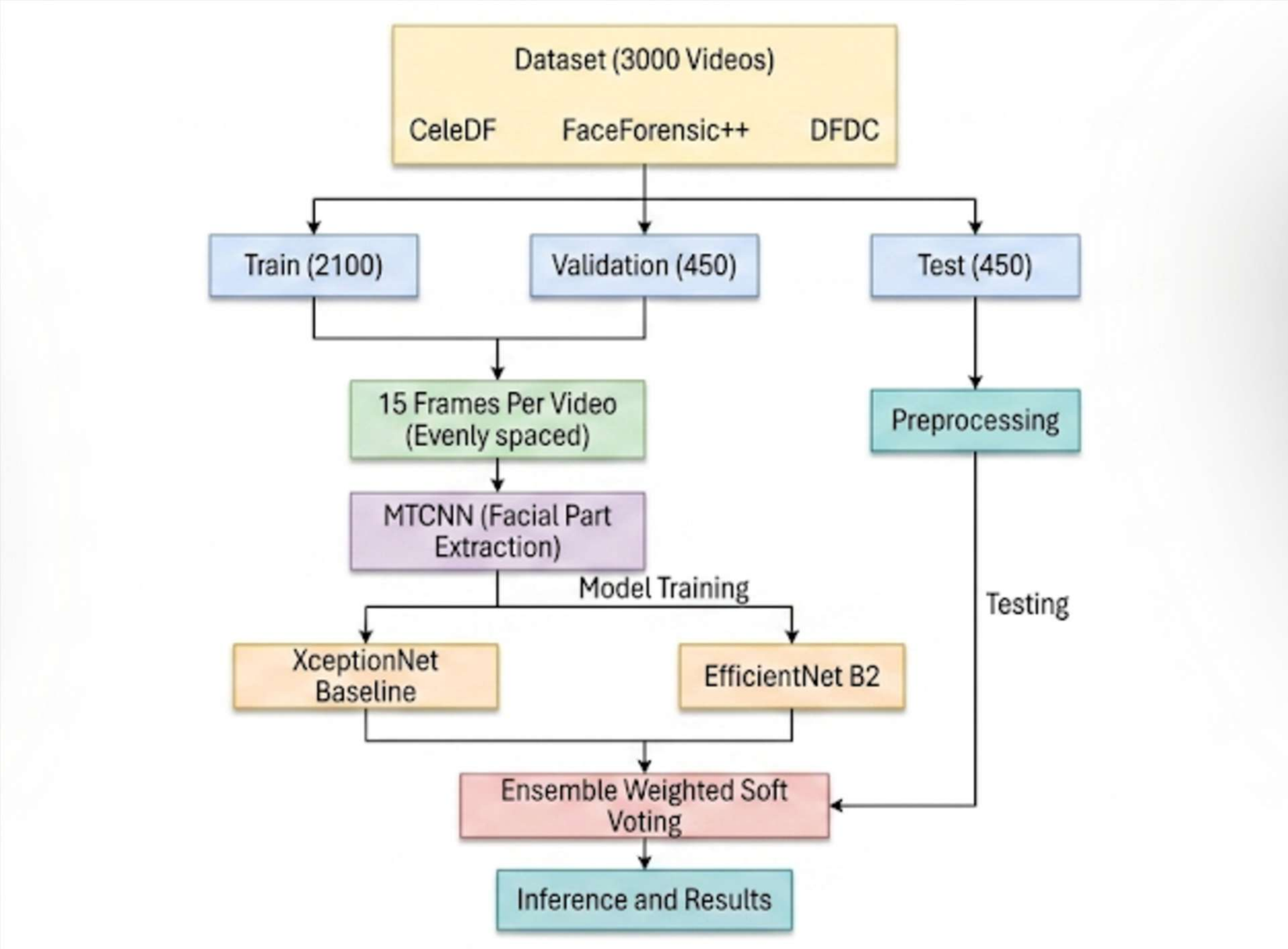
Data Preprocessing

- Extracted 15 frames per video at equal intervals
- Face detection using MTCNN
- Cropping removes background noise
- Data augmentation: flipping, cropping, brightness adjustment

PHASE 1


Dataset
1800 Videos
1400 Videos
200 Videos
200 Videos
Train
Validation
Test
15 Frames Per Video
(evenly spaced)
Pre-Processing
Residual Image
generation
FFT Image
generation
Testing
Processed Data
(Residual + FFT
Images)
Model Training
ResNet18
Classifier
Inference and Result

PHASE 2


Dataset (3000 Videos)
CeleDF
FaceForensic++
DFDC
Train (2100)
Validation (450)
Test (450)
15 Frames Per Video (Evenly spaced)
Preprocessing
MTCNN (Facial Part Extraction)
Model Training
Testing
XceptionNet Baseline
EfficientNet B2
Ensemble Weighted Soft Voting
Inference and Results

Hardware Specifications

COMPONENT	SPECIFICATION
Processor	Intel Core i7
RAM	16 GB
GPU Model	NVIDIA GPU with CUDA Support
GPU Model	NVIDIA RTX 2060
GPU Memory	8 GB VRAM
Storage	100 GB

Software Specifications

SOFTWARE/TOOL	DESCRIPTION
Operating System	Windows
Programming Language	Python 3.8+
Deep Learning Framework	PyTorch
Image Processing Libraries	OpenCV, PIL
Video Processing Tools	OpenCV, FFmpeg
Data Handling	Data Handling
Evaluation Metrics	Scikit-learn
Deployment Framework	Flask

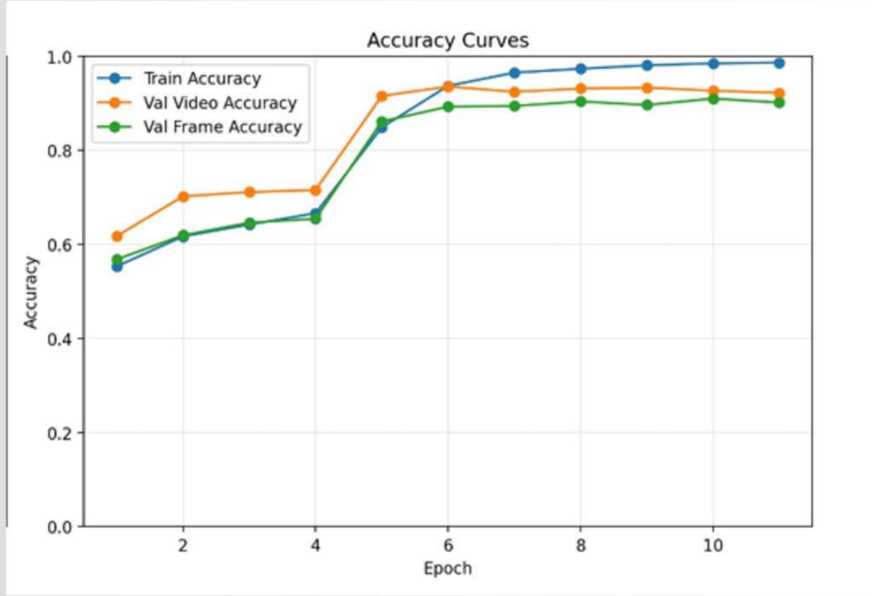
Results

Phase 1 – Camera captured vs Non-camer (AI Video)

- The system successfully classifies videos into Real, AI-Generated, and Deepfake categories
- High frame-level accuracy of 98% achieved using spatial and frequency-based features
- Video-level accuracy further improved by aggregating frame predictions upto 99.5%

Phase 2 – Pristine (Real) vs Deepfake (Manipulated)

- XceptionNet baseline standalone achived 95.5% accuracy.
- EfficiantNetB2 standlone achived around 94%.
- The ensemble model (XceptionNet + EfficientNetB2) achieved the highest performance (~96.67%)
- Slightly lower accuracy in DFDC dataset due to higher realism and complexity



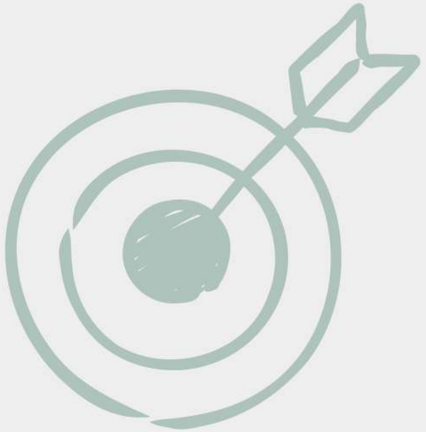
Accuracy curves comparing training, validation, and frame-level performance across epochs for Phase 2 models.

Training vs Validation Accuracy showing stable convergence and high frame-level performance in Phase 1

Results

Level	Accuracy	Precision	Recall	F1 Score
Video-Level	0.9950	0.9901	1.0000	0.9950
Frame-Level	0.9823	0.9800	0.9800	0.9800

Video vs Frame Level Model Performance Metrics in phase 1 AI Video detection



Model	Overall Accuracy	CelebDF	DFDC	FaceForensics++
Xception Baseline	95.56	96.25	89.23	100.00
EfficientNet-B2	94.00	93.13	87.69	100.00
Ensemble	96.67	96.88	92.31	100.00

Overall and Dataset-wise Test Accuracy Comparison in phase 2 manipulated video detection

Conclusion

The project presents a two-phase deep learning framework for detecting both AI-generated and deepfake videos with improved accuracy and reliability. By combining spatial features through residual analysis and frequency-based features using FFT, the system is able to capture subtle inconsistencies that are often missed by traditional methods. The use of multiple benchmark datasets such as DFDC, CelebDF, and FaceForensics++ enhances the model's ability to generalize across different types of manipulations and real-world conditions. Additionally, the integration of ensemble models like XceptionNet and EfficientNet strengthens overall performance and robustness. The system effectively handles variations such as compression, lighting, and video quality, making it suitable for practical applications. Overall, this approach provides a scalable and efficient solution for ensuring video authenticity in the growing landscape of AI-generated media.

Future Enhancements

Include AI-generated videos from different generators such as Sora, Runway, Pika, Veo, and SDXL to make Phase 1 more generalized like Phase 2.
Improve robustness against video compression, low-light conditions, blur, noise, and low-resolution inputs.
Classify different manipulation types separately, such as FaceSwap, Face2Face, FaceShifter, and DeepFakes, instead of only binary detection.
Integrate explainable AI methods to highlight manipulated regions and improve transparency in prediction.
Optimize the model for faster inference so that the system can support real-time or near real-time video verification.

Outcome

- Developed a two-phase deepfake detection system for structured analysis
- Achieved high accuracy using ensemble models (XceptionNet + EfficientNet)
- Improved cross-dataset generalization using multiple benchmark datasets
- Successfully detected AI-generated, real, and deepfake videos
- Enhanced detection using spatial (residual) and frequency (FFT) features
- Achieved robust performance under compression, lighting, and quality variations
- Improved video-level accuracy through frame aggregation techniques
- Built a scalable, efficient, and modular framework
- Reduced misclassification by combining multiple feature representations
- Provided a practical solution for real-world deepfake detection applications

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